

sapp is the top paid app on the Apple App Store in most countries) is seen as being nearly ubiquitous, but failed to gain significant traction in North America. Kik messenger, another free, cross platform app is popular in the US, while Line and Kakao Talk have established themselves in Japan and South Korea respectively.

In order to truly become universal, apps will have to either develop and implement an open standard (something like XMPP), or find a way to integrate with SMS in the way Apple's iMessage has. iMessage is unique in its simplicity and ingenuity – if the recipient is an iMessage user, the message is sent over the internet using iCloud, if not, it's sent as regular SMS / MMS. In addition, the service integrates with the Notification Center on Mac OS X, so users can receive and reply to messages from any Apple device. Many other services have followed suit.

Communication apps are subject to strong network effects, and any newcomers will be faced with an uphill battle of convincing existing users to switch. (Google+ anyone?). Ideally, we'd like all the different providers agree on an open standard (like IMAP is for Email, or even SMS itself), that will create a level playing field. The reality, however, is that any attempts at creating such a standard would be futile, with each company wanting to protect its own business interests. Furthermore,



Developing an open standard for mobile messaging is an uphill task.

attempts at creating a standard might result in a protocol that while being open, doesn't receive much support from existing apps due to the lack of any incentive to switch. The silver lining though, is that going forward, messaging is only going to become cheaper, with greater sharing features and will span a broad spectrum of devices.



Vic Gundotra unveiling Google's new Hangouts app at Google I/O 2013.

RISC chips

If the past few years have seen tremendous growth in the mobile sector, it is chiefly due to the development of low-power and high-performance microprocessors relying on the RISC (Reduced Instruction Set Computing) architecture. An Instruction Set is the complete set of commands or 'instructions' that a microprocessor can follow, and RISC is a form of CPU design which believes that using fewer

total instructions will lead to better, more efficient processors. Leading the path, was a British Semiconductor giant, ARM (Advanced RISC Machines), which came to design a number of CPU designs such as the ARM v11, and the ARM Cortex series. While ARM doesn't manufacture the chips themselves, they license out their designs to companies like Samsung and Qualcomm. The success of the ARM architecture, and of the mobile sector in general is evident from a report that puts Qualcomm and Samsung ahead of AMD in processor sales (which has traditionally held the second position behind Intel)

Traditionally, Intel had held the lead in microprocessor sales, owing to their superior design and state-of-the-art, highly optimized manufacturing process, and had led to Intel capturing the lion's share of the PC segment. However, with PC sales plummeting, Intel has been forced to rethink its strategy, opting to refresh its Atom line of processors (which faced a lot of flak for being



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